

Repair Manuals

Timing chain: removal/installation

Warnings and recommendations

Unless otherwise advised by the manufacturer, the following procedures are recommended:

Always renew the timing chain

If the timing chain is to be re-used, mark the running direction before removal

Never turn the engine in the direction opposite to that of normal operation

Note: Avoid bending, torsion or traction

Always use new gaskets

General

Before disconnecting the battery cable, check the audio system security code

Clean off the old sealant

Apply sealant

Renew the seals

Removal

Disconnect the battery

Remove the inlet silencer

Remove the bulkhead cover

Remove the bulkhead cover insulation

Remove the wiring loom from the cylinder head

Remove the valve cover

Remove the spark plugs

Remove the coolant pump friction wheel

Remove the ancillary drive belt

Remove the crankshaft pulley

Turn the crankshaft clockwise

Vehicles with automatic transmission:

Use a special tool to lock the crankshaft (2 299 362)

Vehicles with manual gearbox:

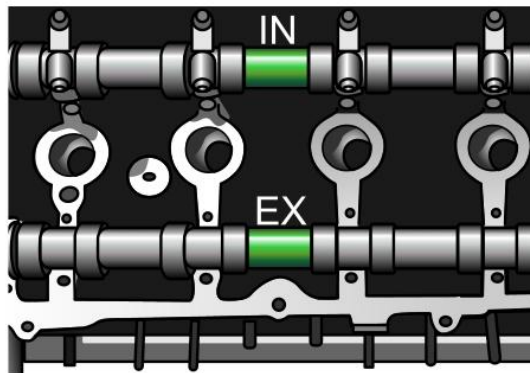
Use a special tool to lock the crankshaft (0 496 709)

All vehicles:

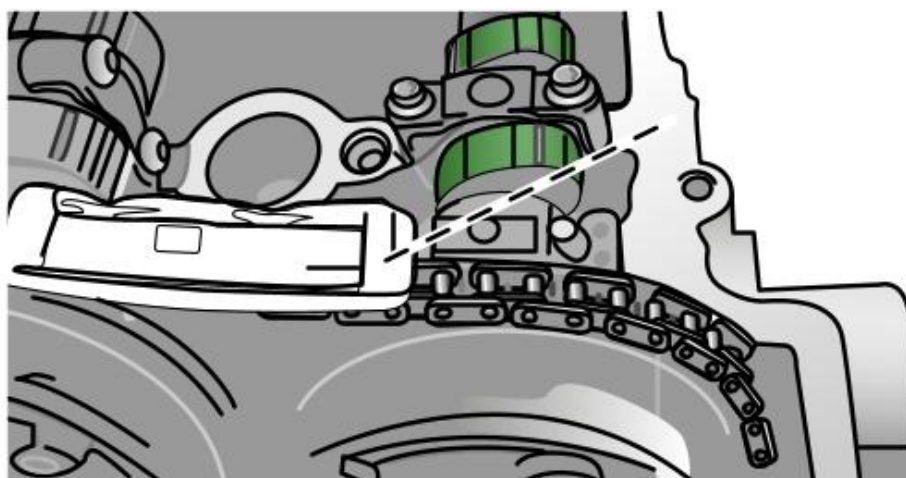
Variable valve timing actuator:

Turn the camshafts to check that the camshaft adjustment units can be locked

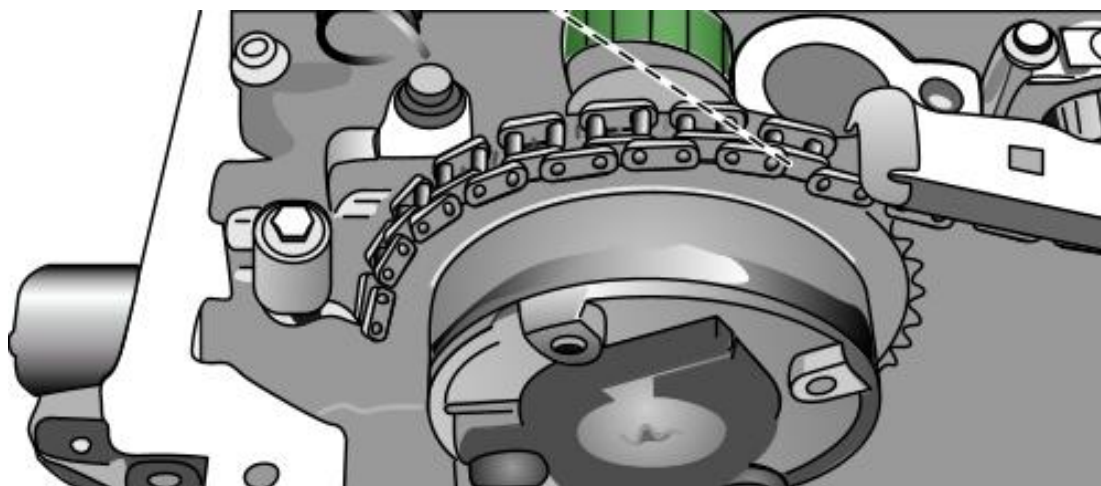
Check the camshaft timing marks (IN, EX)



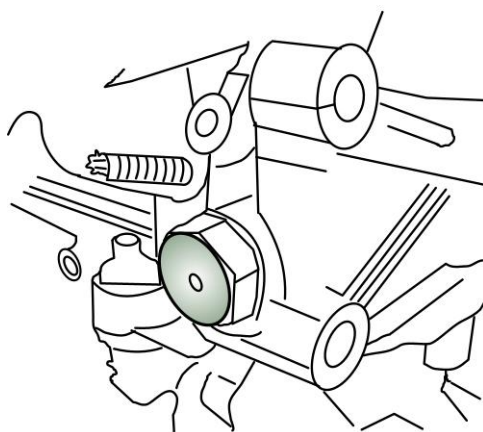
Exhaust camshaft cam angle

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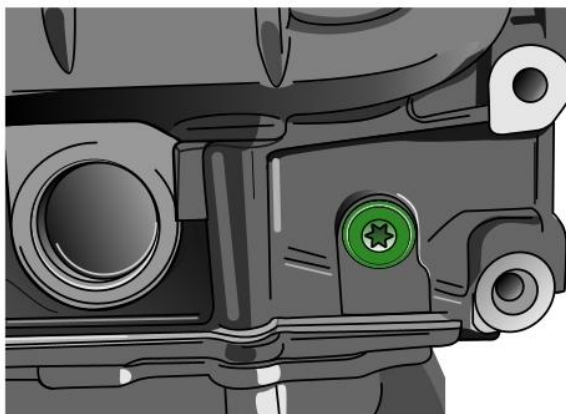
Inlet camshaft cam angle



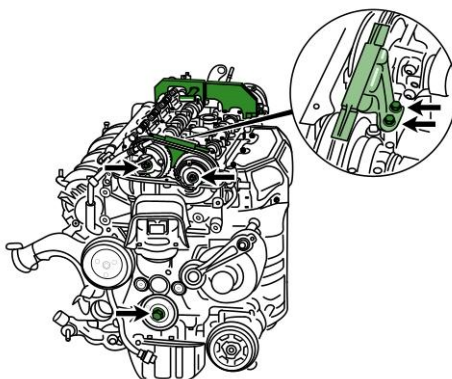
Note: Both camshafts must point upwards at an angle
Use the special tools to lock the camshafts (11 7 440)
Remove the chain tensioner piston



Chain guide rail:
Remove the bolt

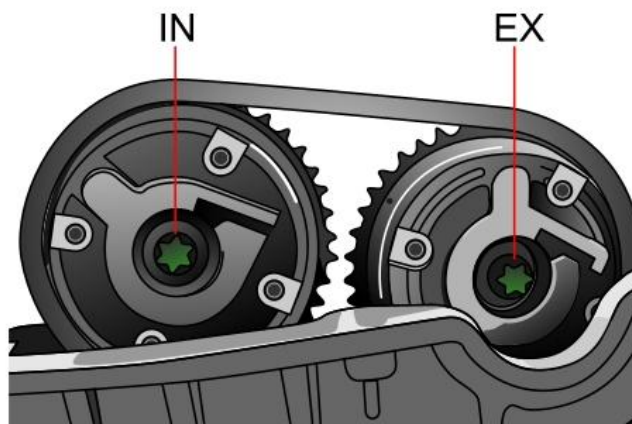
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Remove the timing chain upper guide



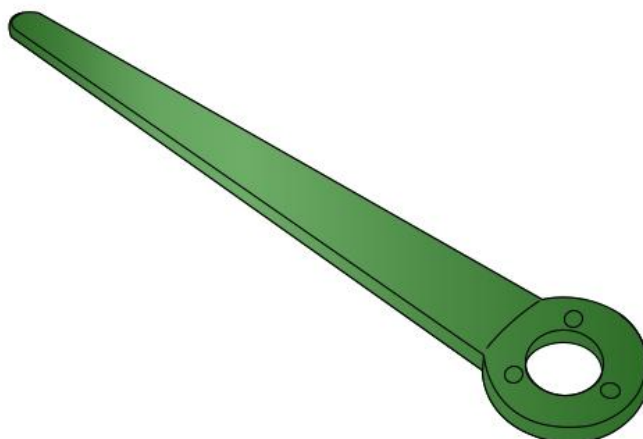
Remove the inlet camshaft timing adjuster (IN)

Remove the exhaust camshaft timing adjuster (EX)



Remove the crankshaft gearwheel centre bolt

Use the special tool to counterhold (11 9 280)

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Bearing pins for guide and tensioner rails:

Remove the bolts



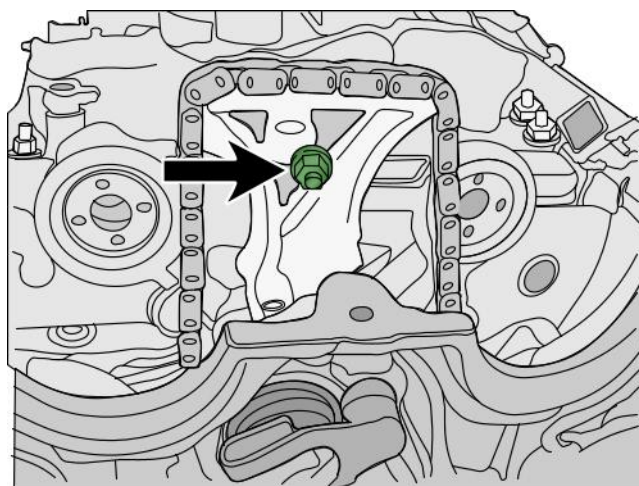
Remove the crankshaft hub

Remove the chain and the crankshaft gearwheel

Remove the chain guide with the timing chain, guide rail and tensioner rail

Note: Check that the oil pump gearwheel and chain are in place

Remove the timing chain guide bolts from the cylinder head (x 2)



Remove the chain guide with the timing chain, guide rail and tensioner rail

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Installation

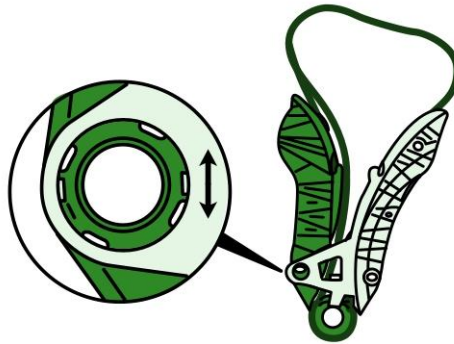
Timing chain assembly:

Fit the chain tensioner guide

Fit the chain guide

Fit the timing chain

Fit the crankshaft gearwheel to the timing chain

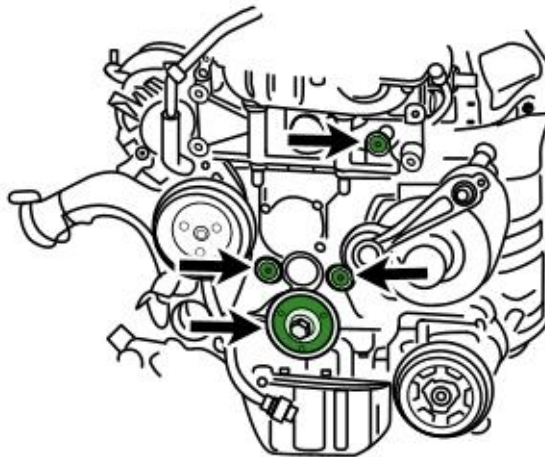


Fit the timing chain assembly from above the cylinder head

Note: Check that the oil pump gearwheel and chain are in place

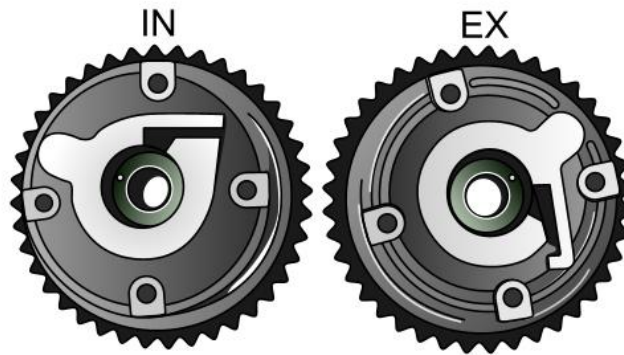
Refit the timing chain rail bolts

Refit the crankshaft hub



VANOS unit:

Check the marks (IN, EX)

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Refit the exhaust camshaft timing adjuster (EX)

Do not fully tighten

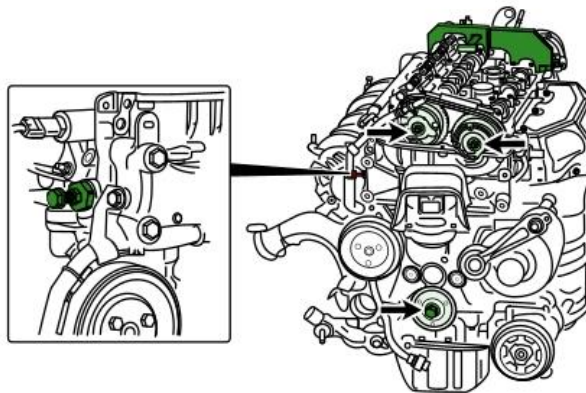
Refit the inlet camshaft timing adjuster (IN)

Do not fully tighten

Refit the timing chain upper guide

Fit the dummy chain tensioner (11 9 340)

Pre-tension the timing chain (0.6 Nm)



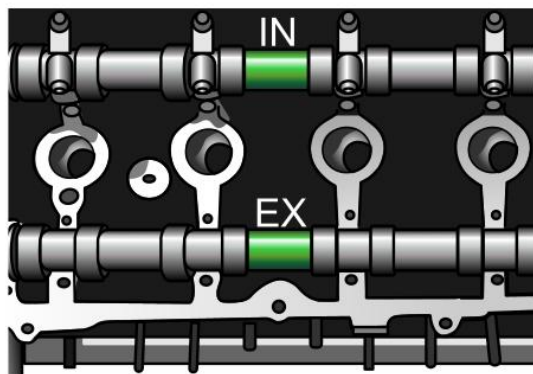
Tighten the inlet camshaft gearwheel

Tighten the exhaust camshaft gearwheel

Remove the dummy chain tensioner (11 9 340)

Refit the chain tensioner piston

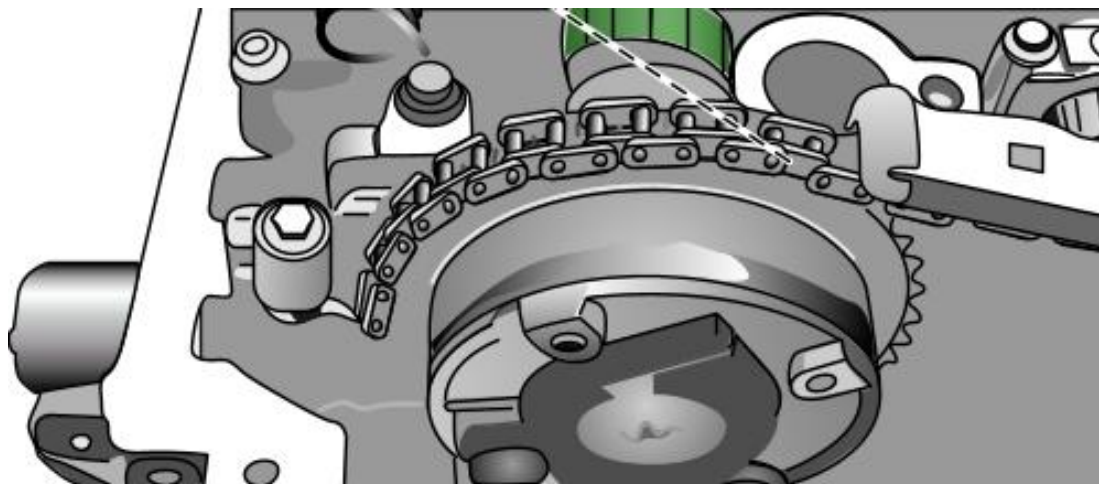
Check the camshaft timing marks (IN, EX)

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Exhaust camshaft cam angle



Inlet camshaft cam angle



Note: Both camshafts must point upwards at an angle

Further refitting is in the reverse order of removal

Torque settings

Valve cover

Camshaft gearwheel

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Timing chain tensioner

Crankshaft pulley

Crankshaft gearwheel

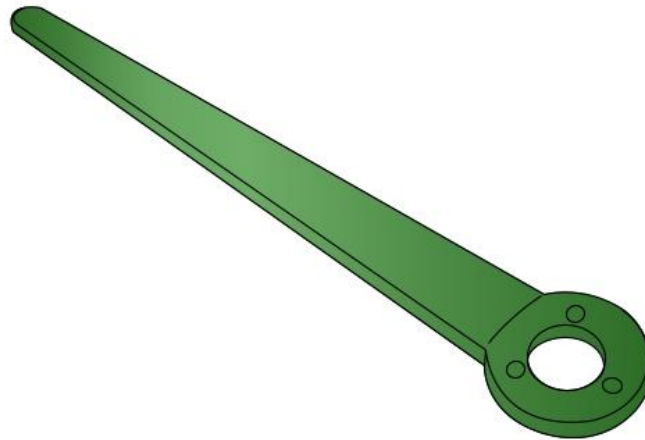
Timing chain guide

Spark plugs

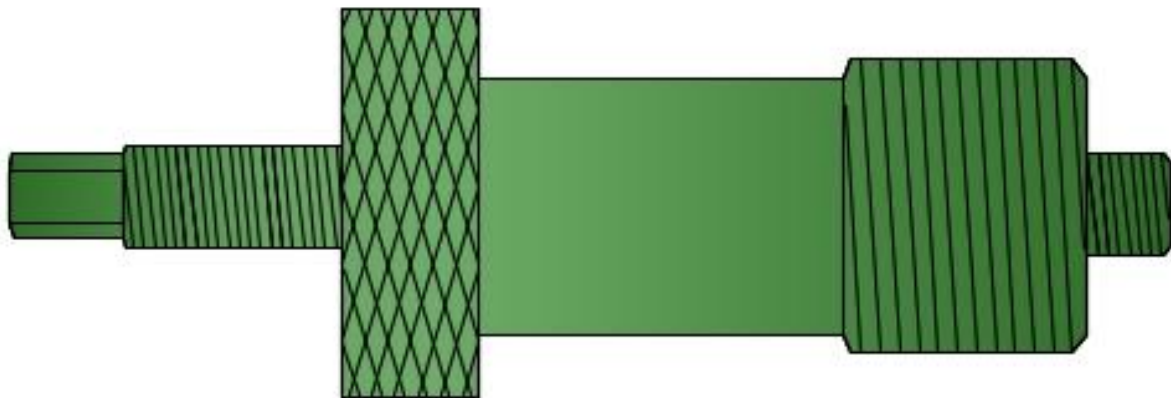
Coolant pump friction wheel (M6: 10 Nm)

Special tools

Crankshaft centering tool: (11 9 280)

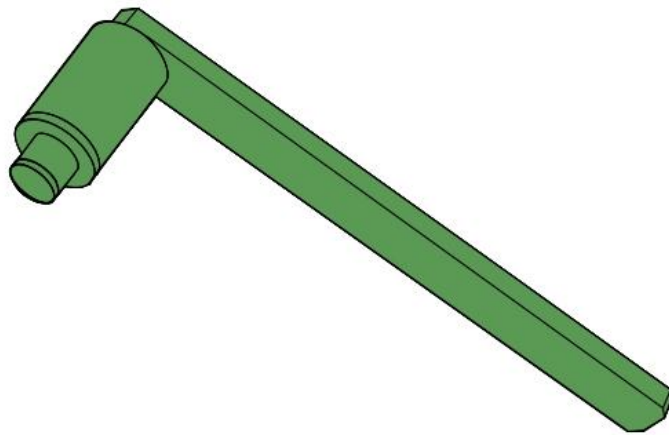


Adaptor: (11 9 340)



Flywheel locking tool: (0 496 709)

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Flywheel locking tool: (2 299 362)

